



ALLEN INSTITUTE *for*
BRAIN SCIENCE

Transcriptomics: Brain to Sequence

Describe Your Neurons Workshop, April 6th 2026

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Lydia Ng

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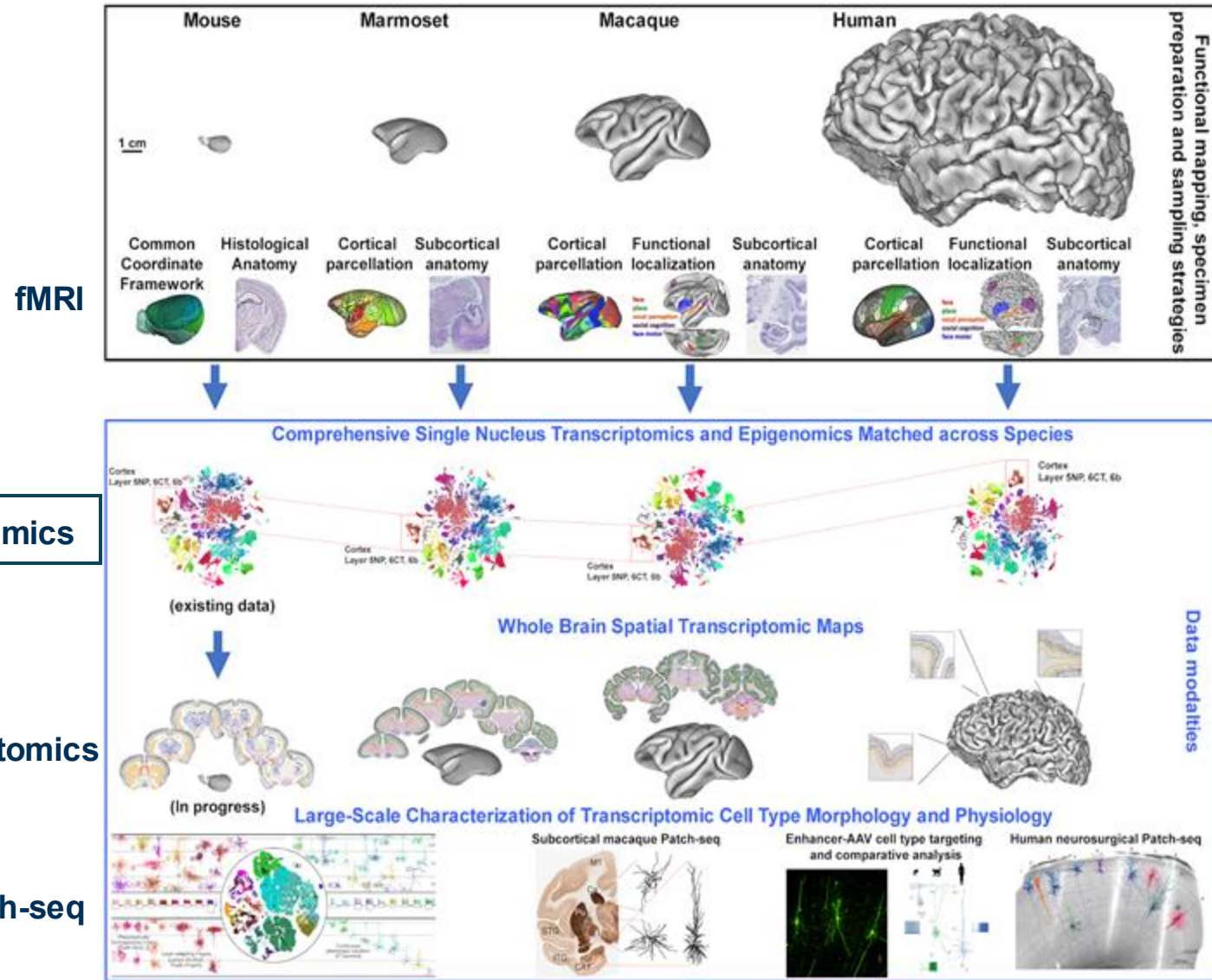
Aaron Garcia

Tamara Casper

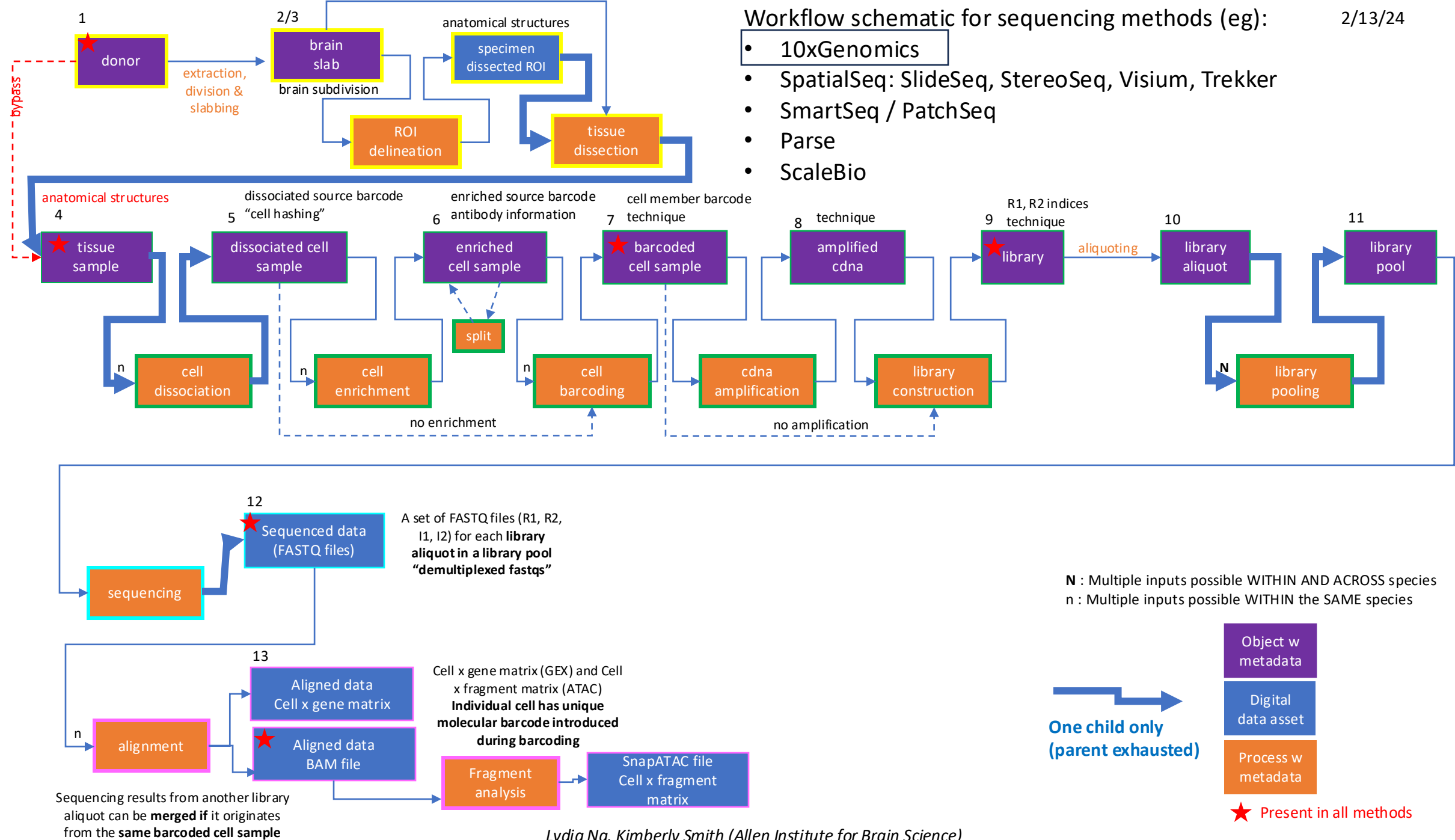
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Cindy van Velthoven

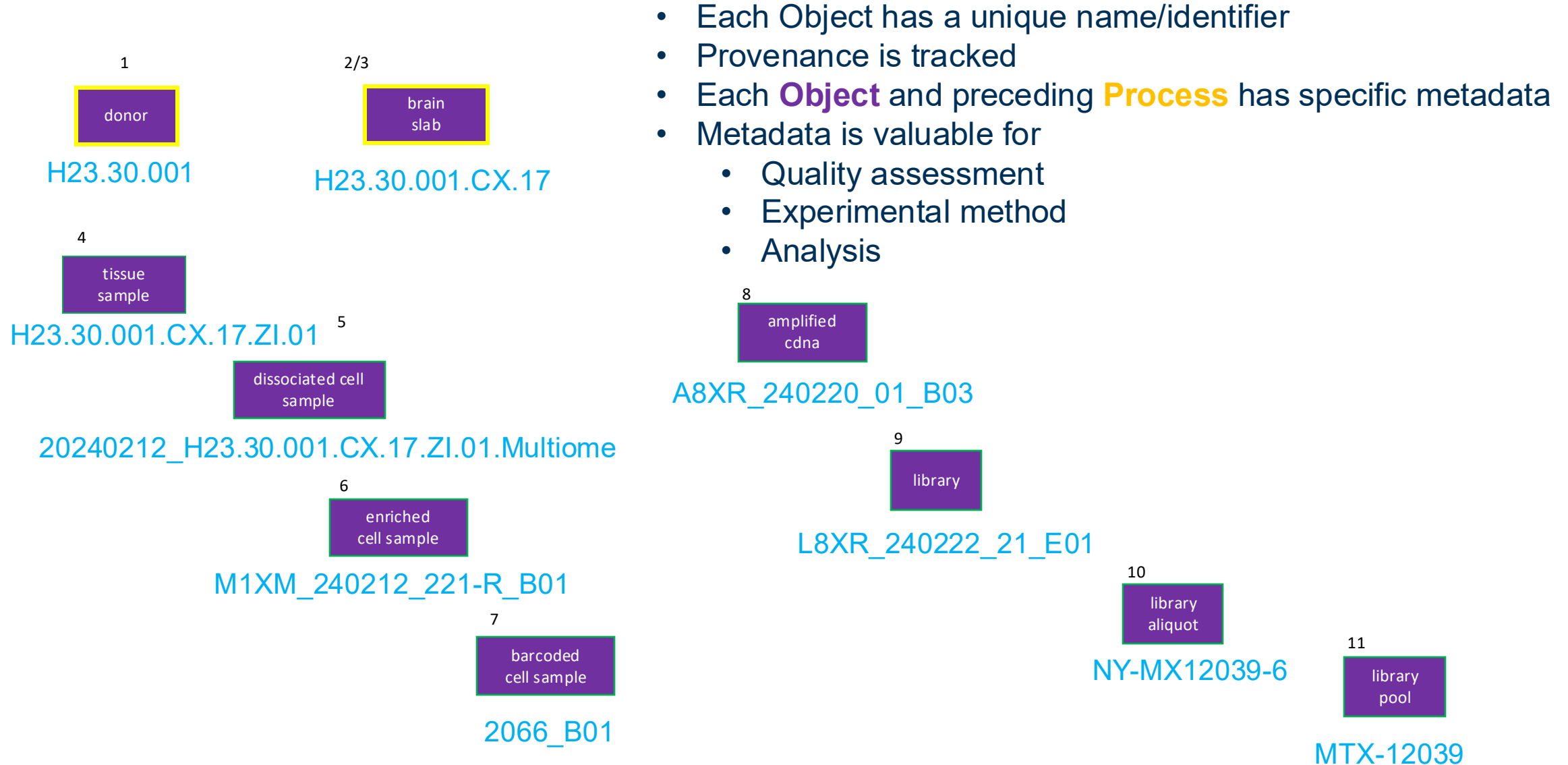
Using transcriptomics to create adult whole brain cell atlases in human and non-human primates



- 10xGenomics
- SpatialSeq: SlideSeq, StereoSeq, Visium, Trekker
- SmartSeq / PatchSeq
- Parse
- ScaleBio



Named Objects – tracking a sample from donor to sequencing library



Centralized information storage for specimen metadata:

Neuroanatomy-anchored Information Management Platform for Collaborative BICAN Data Generation (NIMP)

- Specimen Portal
- Seq Library Portal
- Donor to Library Aliquot identifier assigned (NHashID) and metadata registration
- <https://brain-specimenportal.org/>
- UTHHealth Houston

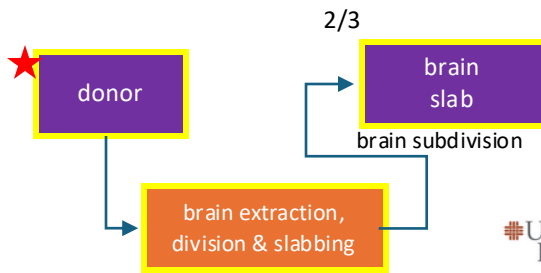
NIMP: Neuroanatomy-anchored Information Management Platform for Collaborative BICAN Data Generation

NIMP is developed under NIH BRAIN Initiative's BICAN U24MH130988 award as a part of the coordinating unit for biostatistics, informatics, and engagement (CUBIE) for the BRAIN Initiative Cell Atlas Network (BICAN) program.

NIMP consists of two portals for BICAN collaborative data generation: the Specimen Portal and the Sequence Library (SeqLib) Portal. The Specimen Portal focuses on tissue management from donors to brain slabs and annotated brain samples. The SeqLib Portal manages the workflow starting from tissue, all the way downstream to track data deposition to assay-dependent, data-modality-specific archives. Both portals work in tandem to generate multimodal genomic data that can be traced back to their anatomical origins using the Allen Brain Atlas.

The portals provide multiple types of data interfaces through dashboards, APIs, faceted queries, and batch data ingestion and exporting. All of the underlying functionalities are achieved through a robust agile development strategy using NHash resource identifiers, metadata standardization, active combinatorial dashboarding, resource provenance linkage and rendering (e.g. Sankey diagrams), and dedicated interfaces with NIH Neuro Biobank, sequencing centers, NeMO, and the larger BICAN data ecosystem.

[Sign Up](#)[Log In](#)[FAQ](#)[Architectural Diagrams](#)



Transcriptomics begins with a donor or animal brain

3/29/24
H23.30.001

UTHealth
Houston Specimen Portal

Home Projects Seq Library Portal Resource Explorer Pinning Leaderboard

Donor

Slab and ROI

Request

+ Donor

Donors via CSV

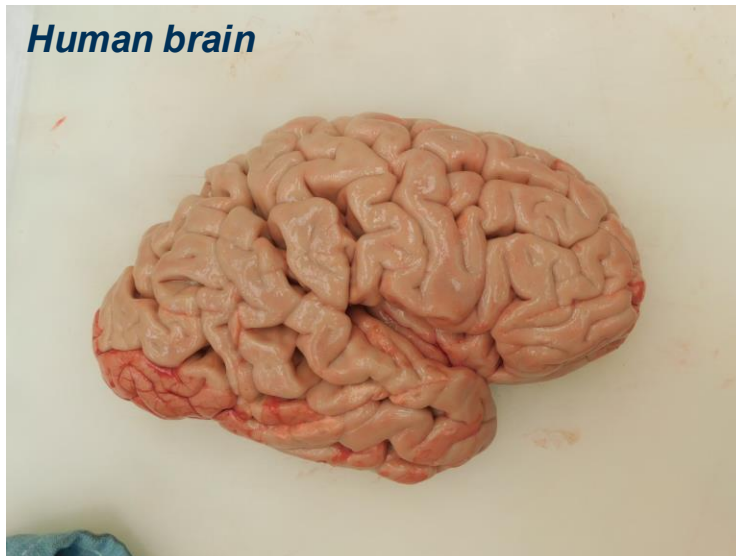
Donor template

Show 5 entries

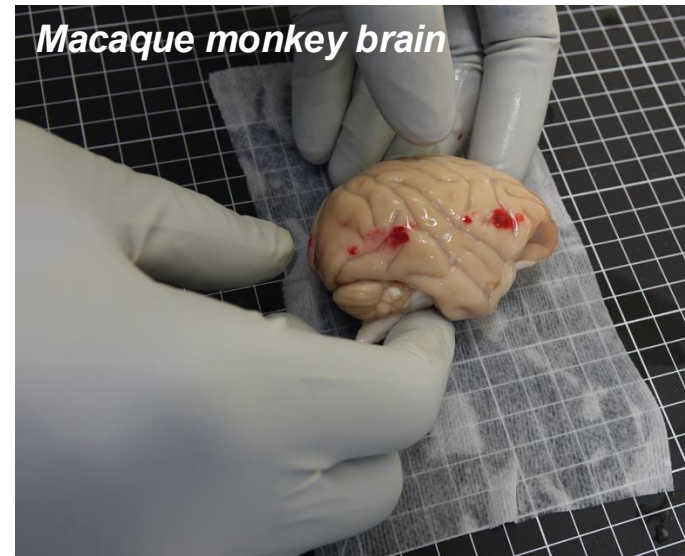
Export current page (or selected rows) as CSV

Subject ID	NHash Donor ID	Species Type	Age at Death	Sex	
H19.30.002	DO-RLQJ8666	NCBITaxon:9606 (Human Adult)	29.0	Male	Q
H23.30.001	DO-CYPH5324	NCBITaxon:9606 (Human Adult)	50.0	Male	Q

Human brain



Macaque monkey brain



Human Cell Types
Aaron Garcia

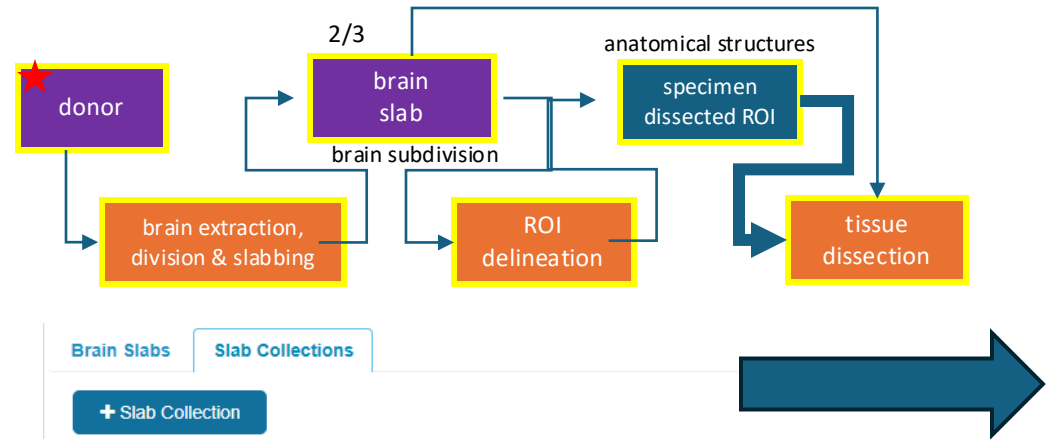
Each donor and animal brain has associated metadata

- Minimal metadata associated with each brain:

- Species
- Sex
- Age:
 - Value
 - Unit
 - Reference Point
 - Description
 - e.g., 6 : months : birth : '6 months'

donor_local_name	age_at_death_value	age_at_death_unit	age_at_death_reference_point	age_at_death_description	donor_species	biological_sex
TB21.39.005	6	months	birth	6 months	NCBITaxon:37347	Male
TB21.39.002	1	years	birth	1 year	NCBITaxon:37347	Female
QM23.50.003	6	years	birth	6 yrs	NCBITaxon:9544	Male

- Human donors will have much more extensive metadata
- Mice have genetic background and potentially viral injection metadata



Brains are first subdivided into slabs

Data Entry Form			
Local Donor ID	H23.30.001	NHash Donor ID	DO-CYPH5324
Brain Subdivision	cerebrum	Hemisphere	left
Visible Slab Face	posterior	Thickness	4.0 mm
*Local Slab ID	UWA-7480.17	NHash Slab ID	SL-CYPHPL532436
<button>Update Brain Slab</button>			

Brain Slabs

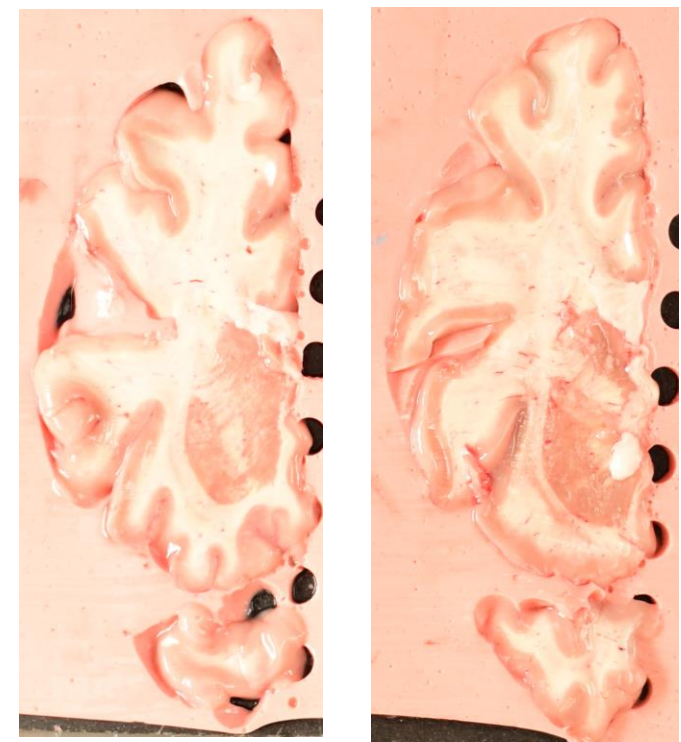
Slab Collections

+ Slab Collection

Brain Subdivision	Hemisphere	Number of Slabs	Slab Thickness	Collection Image
cerebrum	left	2	4.0 mm	
cerebrum	left	2	4.0 mm	

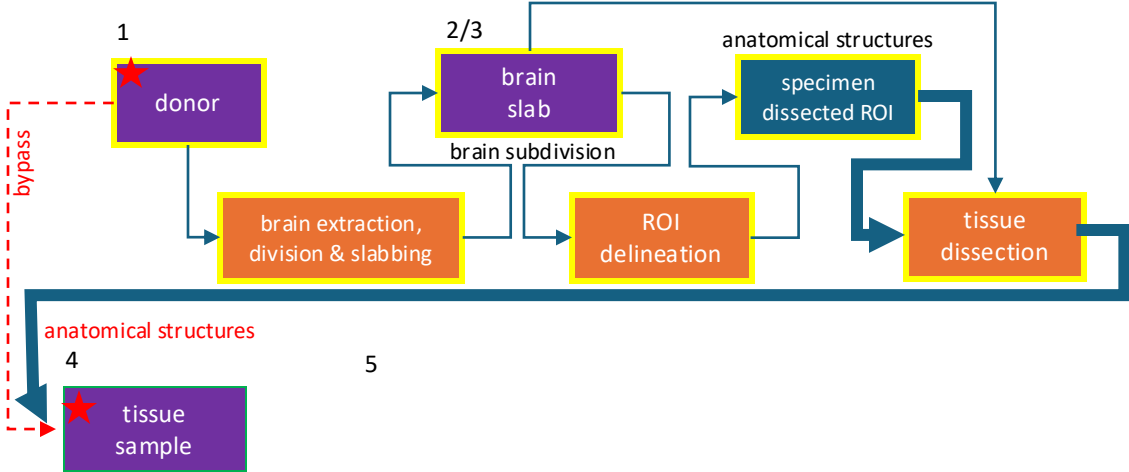


H23.30.001.CX.17



Human Cell Types
Aaron Garcia

Regions of interest (ROIs) are drawn on tissue slabs and pinned to a reference atlas



- Defines area to be dissected
- Area contains 1 or more anatomical structures

Area	Local Name	Status	Pin
	RI-DDZBXB665207 H23.30.001.CX.17.R.01	pre_approved	
	RI-FJEIGD233768 H23.30.001.CX.17.VA.01	pre_approved	
	RI-ABBHCC431028 H23.30.001.CX.17.ANC.01	pre_approved	
	RI-PMRPSN968979 H23.30.001.CX.17.HTH.01	pre_approved	
	RI-GIEMFD816247 H23.30.001.CX.17.HTH.02	pre_approved	
	RI-EHIFCJ955994 H23.30.001.CX.17.ZI.01	pre_approved	



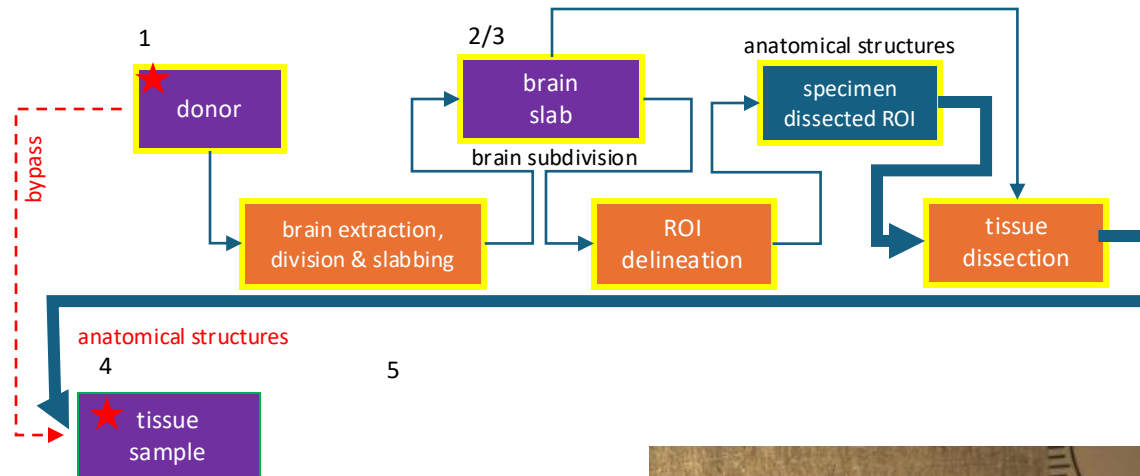
H23.30.001.CX.17.ZI.01
(Zona Inserta)



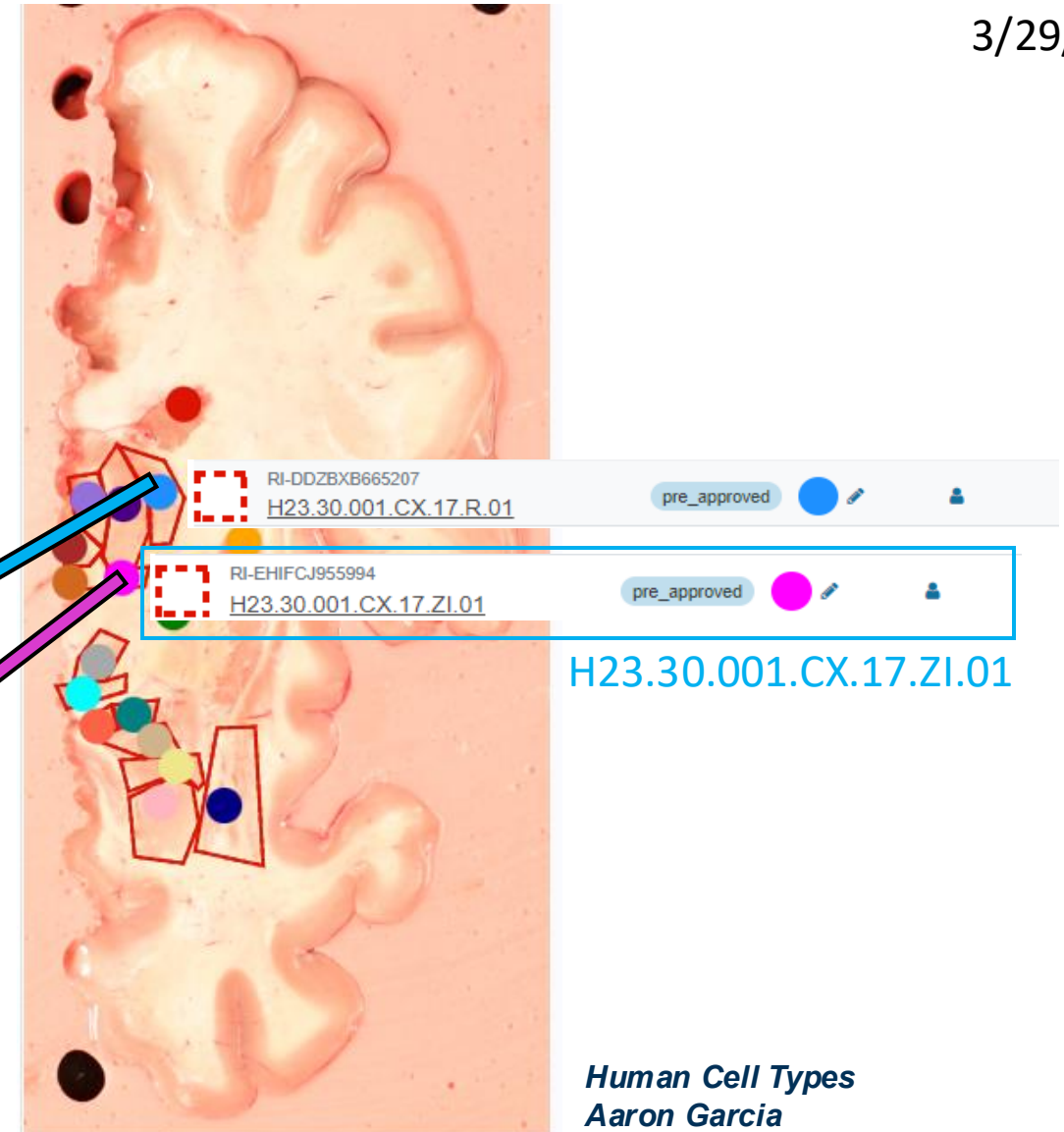
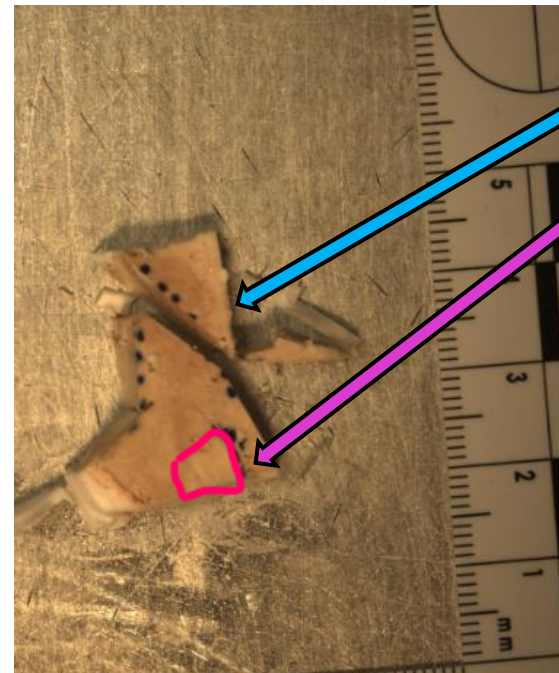
Human Cell Types
Aaron Garcia

ROIs are dissected to produce tissue samples

3/29/24



- Slab is dissected on a cold table to keep the tissue frozen.
- Ink is used to delineate the ROI to be dissected.
- Each tissue piece is placed into a tube and kept frozen until next step.



Each tissue sample has associated metadata

tissue_structure.csv

tissue_sample_local_name	structure
H23.30.001.CX.17.ZI.01_ZI	DHBA:10463



Specimen Portal
Seq Library Portal

Home

Projects

Seq Library Portal

Resource Explorer

Pinning Leaderboard

NIMP Terminology

Table

Tree Structure

Atlas Terminology

zona

Developing Human Brain Atlas

neural plate

neural tube

brain

forebrain (prosencephalon)

gray matter of forebrain

diencephalon

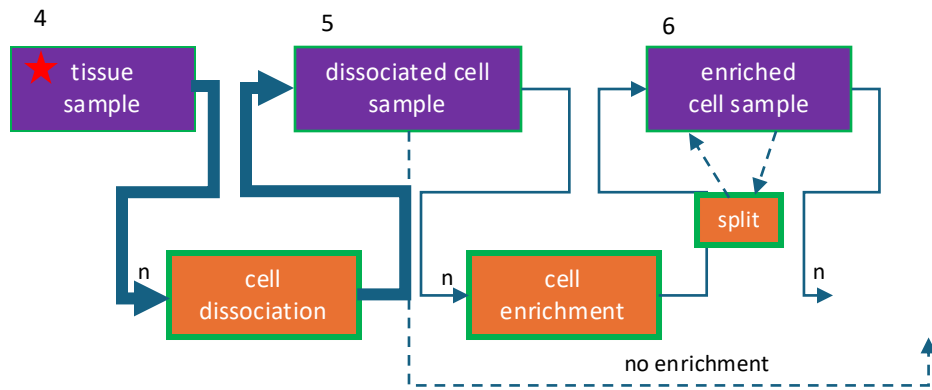
thalamus

ventral thalamus

zona incerta

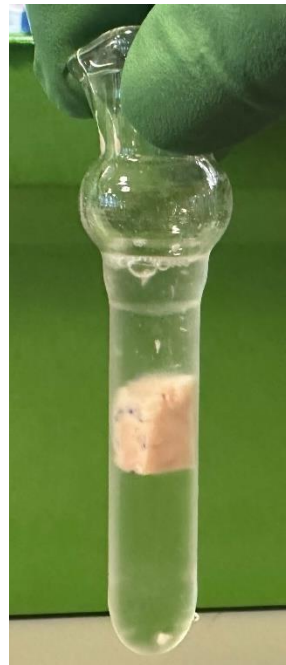
Detailed Information

ID	DHBA:10463
Name	zona incerta
Acronym	ZI
Parent ID	DHBA:10461 (ventral thalamus)
Type	DHBA



Tissue samples are homogenized to create dissociated cell samples

1. Frozen tissue block [H23.30.001.CX.17.ZI.01](#) thaws in buffer.
2. Sample is Dounce homogenized.
3. Myelin removed with a density gradient.
4. Sample is named as unique Dissociated Cell Sample
[20240212_H23.30.001.CX.17.ZI.01.Multiome](#)



1



2



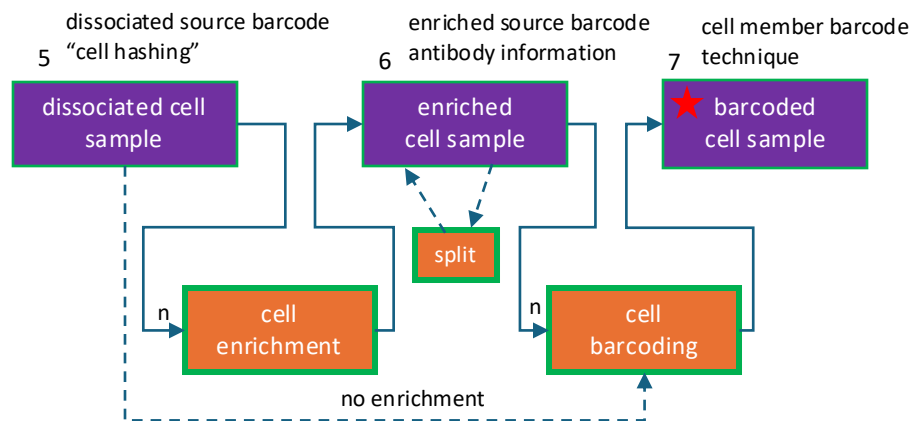
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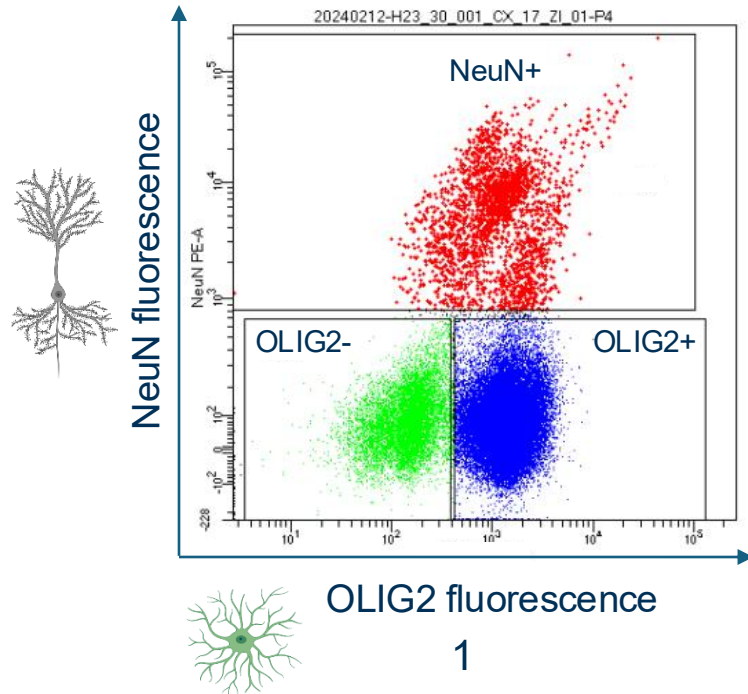
4

Tissue Processing
Tamara Casper

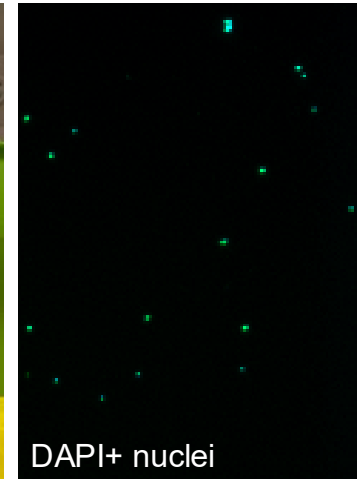
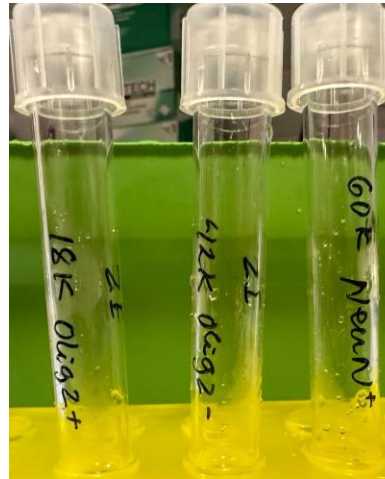
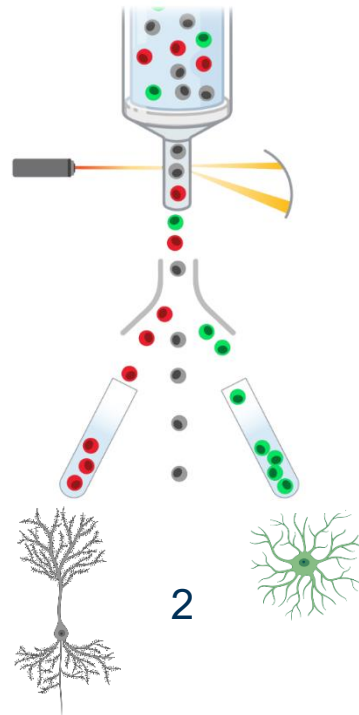
Flow cytometry (FACS) is used to generate an enriched cell sample



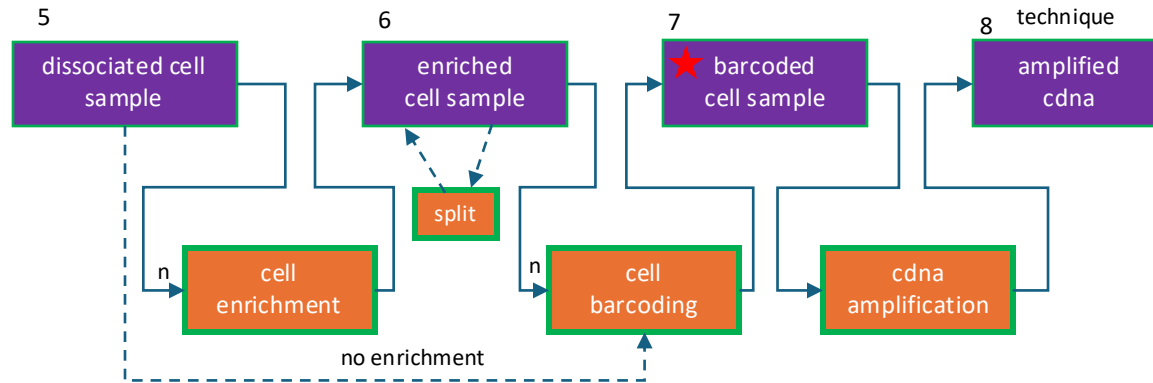
1. Dissociated Cell Sample [20240212_H23.30.001.CX.17.ZI.01.Multiome](#) is enriched using FACS.
2. Target cell populations are collected into FACS tubes.
3. Sorted cells are combined to achieve desired proportion of each population.
4. Cells are counted using a NucleoCounter and DAPI stain.
5. Final sample is handed off as Enriched Cell Sample [M1XM_240212_221-R_B01](#) to load into a 10x Port.



50% NeuN+, 35% OLIG2-, 15% OLIG2+

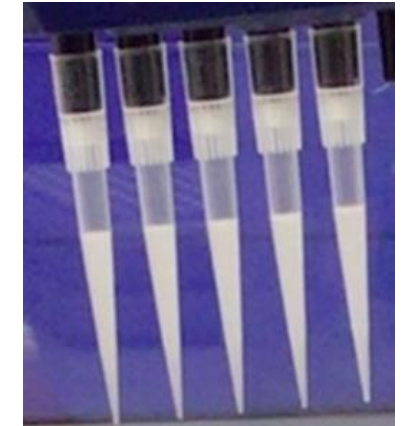
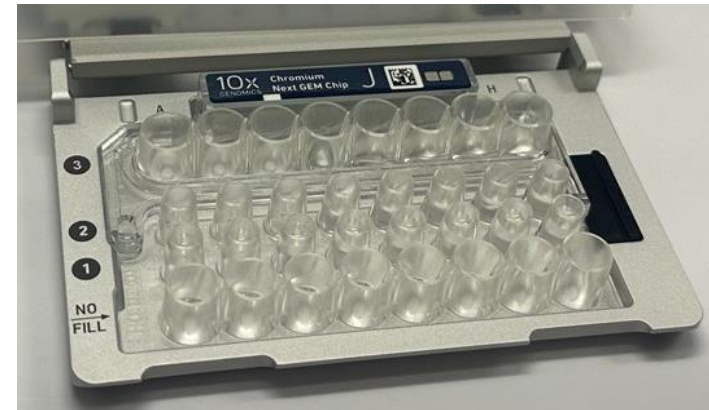
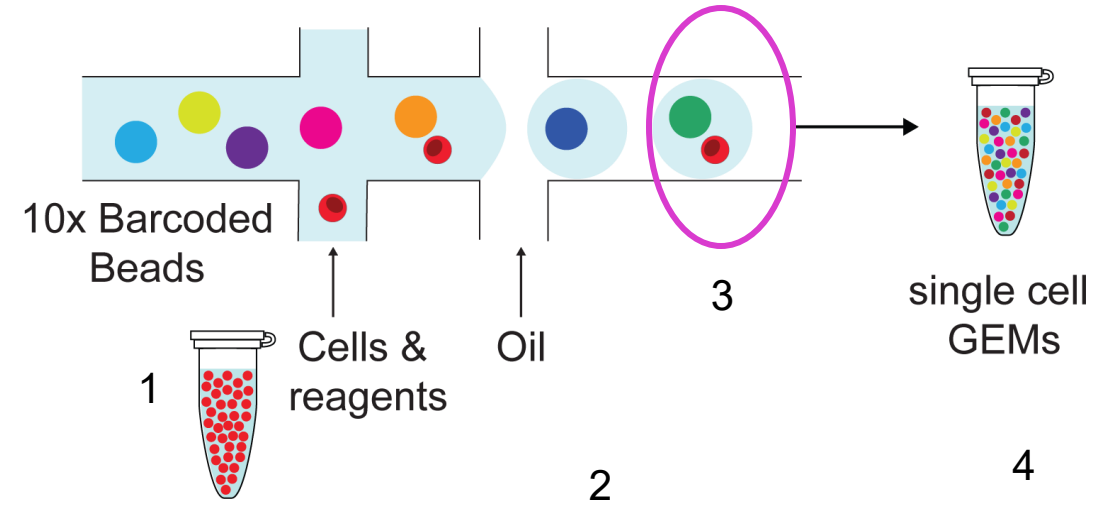


5
Tissue Processing
Tamara Casper



1. Enriched Cell Sample [M1XM_240212_221-R_B01](#) is handed off to the Rseq team, where the cells are added to a reverse transcriptase master mix
2. and loaded into a microfluidic chip along with the 10x barcoded gel beads, and partitioning oil.
3. The chip is run in the Chromium controller where the GEMs* are formed using pressure.
 - At this point each single cell is uniquely barcoded.
4. The GEMs are then harvested from the chip, given Barcoded Cell Sample name [2066_B01](#), and are taken through reverse transcription to convert the mRNA into cDNA.

Each Cell is Uniquely Barcoded within a Barcoded Cell Sample



Unique Cell Member = BarcodedCellSampleName_uniqueBarcode

[2066_B01_AAACAGCCAACCGCCA](#)

Rseq Core
Amy Torkelson

Each cell sample has associated metadata

dissociated_cell_sample.csv

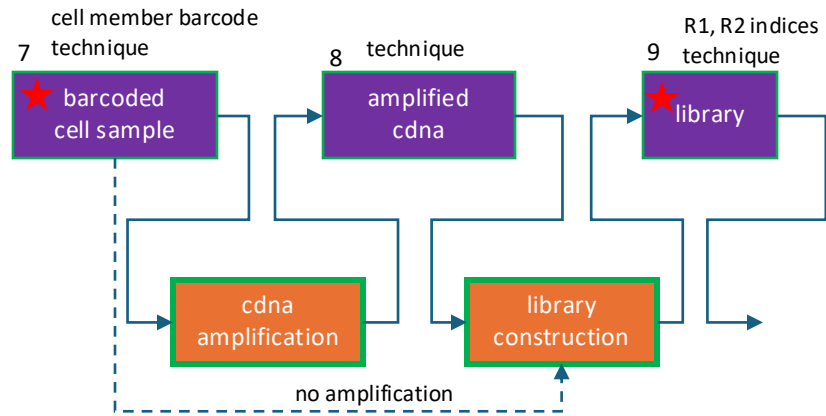
dissociated_cell_sample_local_name	dissociated_cell_sample_cell_label_barcode	dissociated_cell_sample_cell_prep_type	dissociated_cell_sample_preparation_date	project_identifier
20240212_H23.30.001.CX.17.ZI.01.Multiome		Nuclei		UM1MH130981 - Ed Lein

enriched_cell_sample.csv

enriched_cell_sample_local_name	enriched_cell_sample_cell_label_barcode	enriched_cell_sample_preparation_date	project_identifier	enrichment_population	histone_modification_marker
M1XM_240212_221-R_B01		2/12/2024	UM1MH130981 - Ed Lein	50% NeuN+, 35% OLIG2-, 15% OLIG2+	

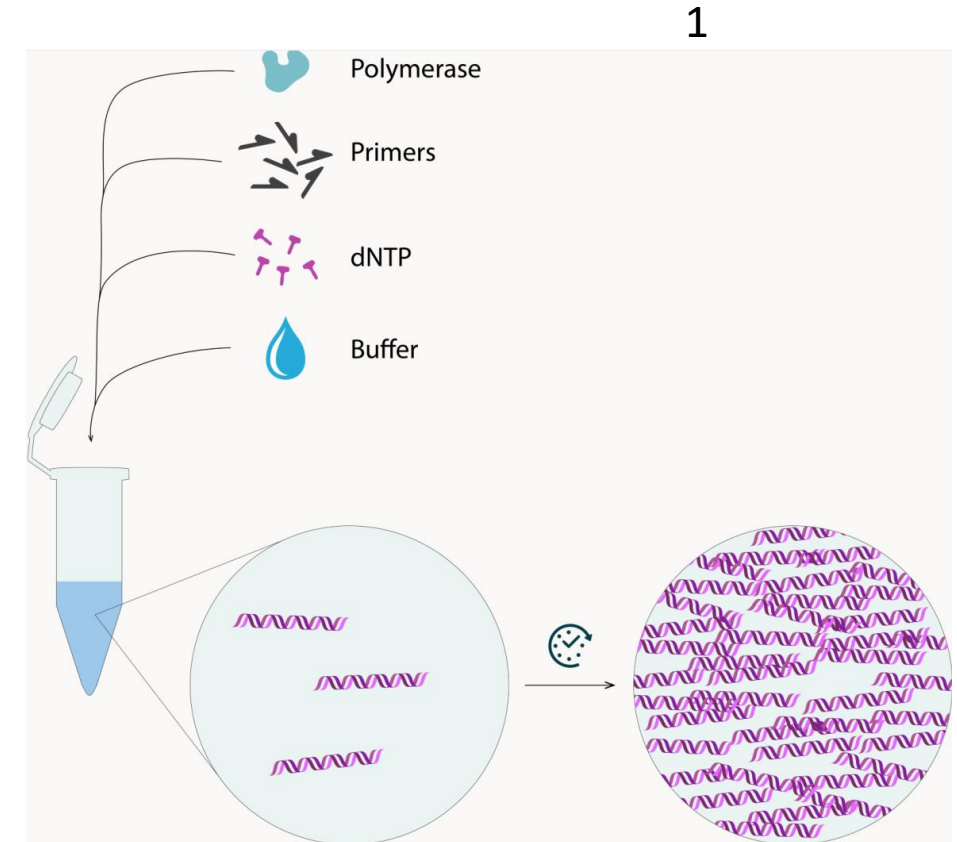
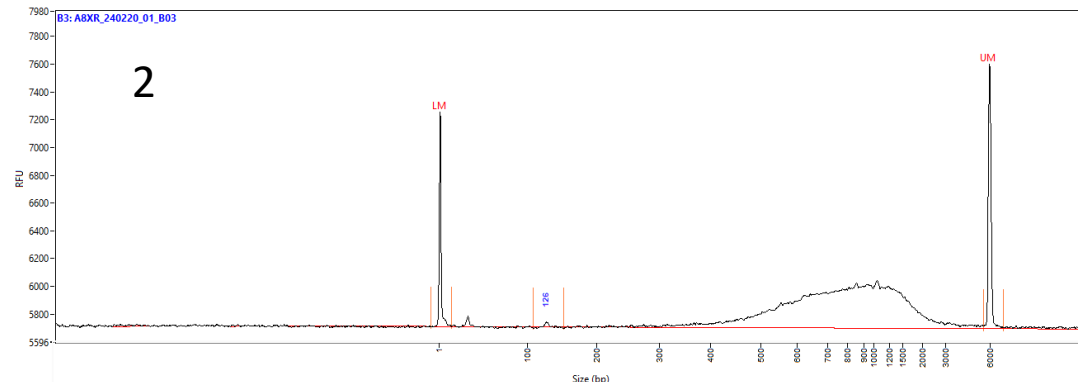
barcoded_cell_sample.csv

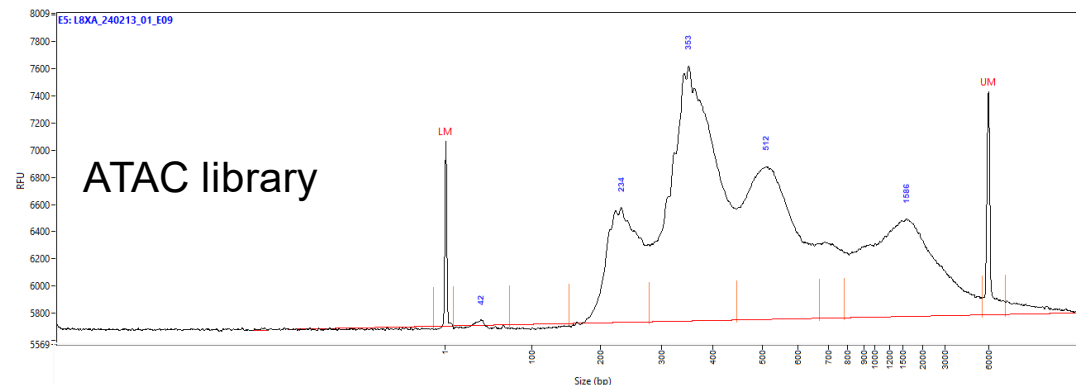
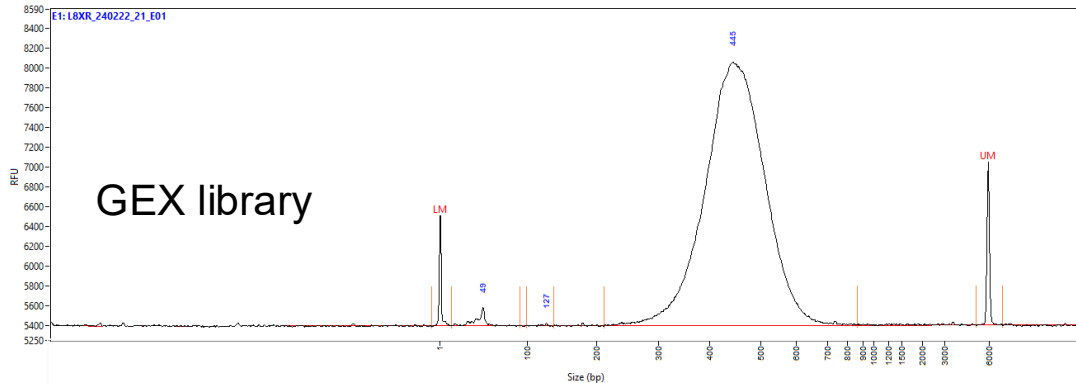
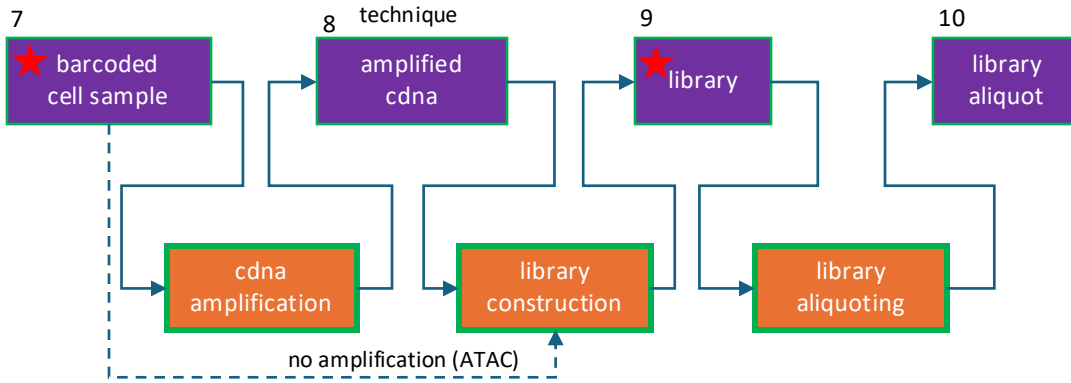
barcoded_cell_sample_local_name	barcoded_cell_sample_source_type	barcoded_cell_sample_number_of_expected_cells	barcoded_cell_sample_preparation_date	project_identifier	barcoded_cell_sample_port_well	barcoded_cell_input_quantity_count	barcoded_cell_sample_technique
2066_B01	enriched_cell_sample	10000		UM1MH130981 - Ed Lein	B01	15876	Multiome



Many copies of each gene in every cell are generated, creating an amplified cDNA product

1. Barcoded Cell Sample [2066_B01](#) goes through cDNA amplification to generate sufficient copies of all gene transcripts expressed in every cell.
2. The cDNA amount and size is determined using the Fragment Analyzer instrument, producing an electropherogram. This is used to Pass/Fail each Amplified cDNA sample.
3. An Amplified cDNA name is assigned: [A8XR_240220_01_B03](#)

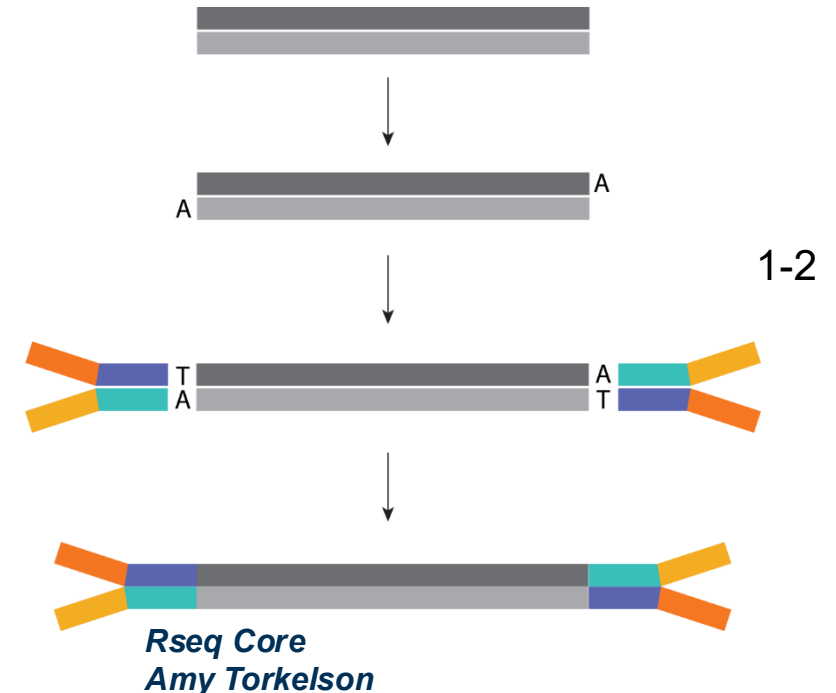




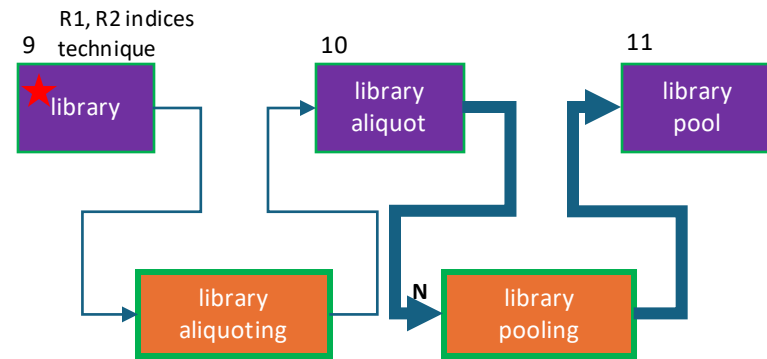
3

Amplified cDNA is then uniquely indexed to generate a Library ^{3/29/24}

- A8XR_240220_01_B03 is then taken through library construction: L8XR_240222_21_E01
 1. cDNA is fragmented to reduce size of molecules for sequencing
 2. Unique indexes are added to each Library to allow for pooling during sequencing
 3. The Library amount and size is determined using the Fragment Analyzer instrument, producing an electropherogram. This is used to Pass/Fail each Library sample.



A portion of each library is combined with other indexed libraries and is ready for sequencing (Aliquots into Pool)

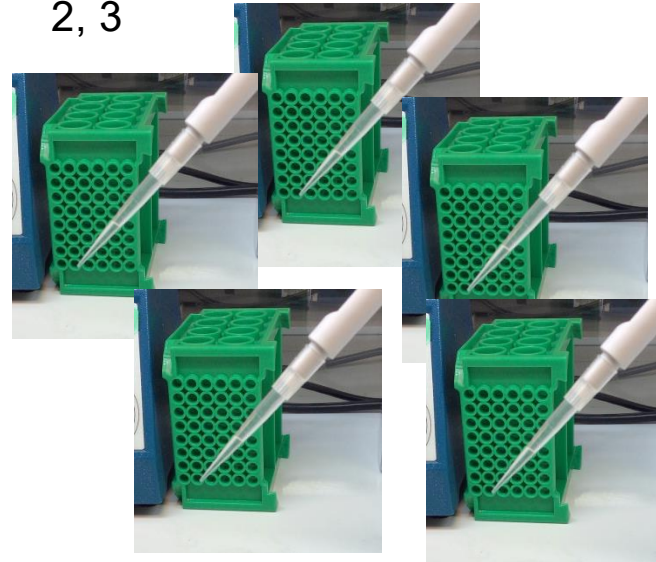


1. To prepare for sequencing, libraries are normalized to 10nM
2. ~30ul of the 10nM library is taken as a Library Aliquot for sequencing [NY-MX12039-6](#)
3. Multiple (5-10) Library Aliquots are combined into a single Library Pool [MTX-12039](#) for sequencing
4. The Indexes allow for separation of sequencing reads to each Library during Demultiplexing

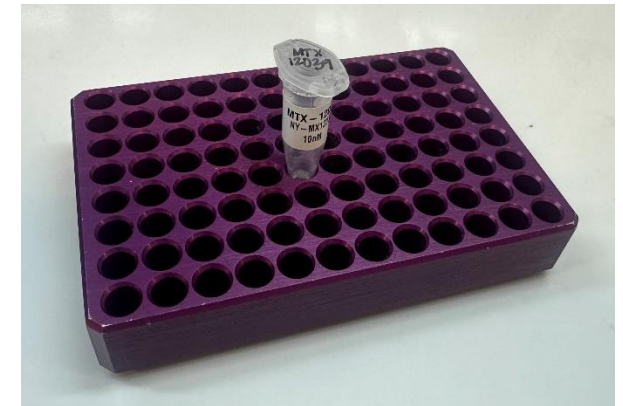
1



2, 3



3



Each amplified cDNA & library has associated metadata

amplified_cdna.csv

amplified_cdna_local_name	barcoded_cell_sample_local_name	amplified_cdna_preparation_date	project_identifier	amplified_cdna_amplified_quantity_ng	amplified_cdna_pcr_cycles	amplified_cdna_rna_amplification_pass_fail	amplified_cdna_percent_cdna_longer_than_400bp	cdna_amplification_set
A8XR_240220_01_B03	2066_B01	2/20/2024	UM1MH130981 - Ed Lein	102.35	10	Pass	0.999	A8XR_240220_01

library.csv

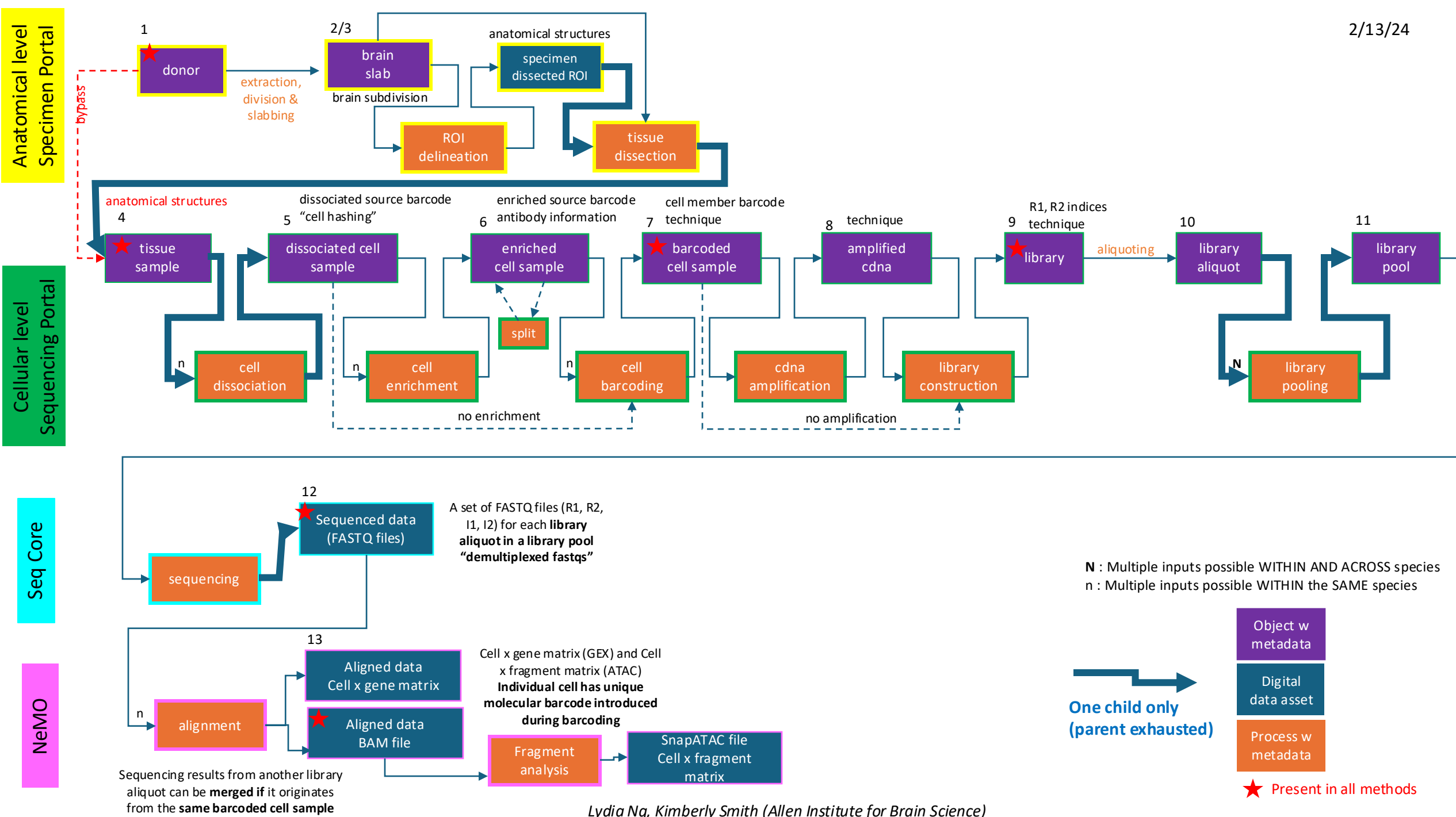
library_local_name	library_technique	library_r1_index	library_preparation_date	library_additional_comments	library_source_type	barcoded_cell_sample_local_name	amplified_cdna_local_name	project_identifier	library_avg_size_bp	library_concentration_nm	library_input_ng	library_prep_pass_fail	library_prep_set	library_quantification_fmol	library_quantification_ng
L8XR_240222_21_E01	10xMultiome;GEX	SI-TT-E1	2/22/2024		amplified_cdna		A8XR_240220_01_B03	UM1MH130981 - Ed Lein	451	278.758	102.35	Pass	L8XR_240222_21	9756.54	2904.13

library_pool.csv

library_pool_local_name	library_pool_local_tube_id	library_pool_tube_barcode	library_pool_preparation_date
MTX-12039	NY-MX12039	K0159735	2/27/2024

library_pool_component.csv

library_pool_local_name	library_local_name	library_aliquot_local_name
MTX-12039	L8XR_240215_01_D09	NY-MX12039-1
MTX-12039	L8XR_240222_21_H01	NY-MX12039-2
MTX-12039	L8XR_240222_21_A02	NY-MX12039-3
MTX-12039	L8XR_240215_01_B09	NY-MX12039-4
MTX-12039	L8XR_240215_01_F08	NY-MX12039-5
MTX-12039	L8XR_240222_21_E01	NY-MX12039-6
MTX-12039	L8XR_240215_01_C09	NY-MX12039-7
MTX-12039	L8XR_240222_01_D10	NY-MX12039-8
MTX-12039	L8XR_240222_01_D12	NY-MX12039-9



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Amber Getman
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